

Tentative title: Statistical mechanics of bifurcations explain phase transitions in finite size disordered systems

Birth of sequential memory recall in delayed non-reciprocal Hopfield networks

Abstract

A disordered system of finite size is not merely a noisy image of its thermodynamic limit: a single sample can carry an internal statistical ensemble of local bifurcations, and different collective phenomena can be governed by different order statistics of that ensemble. We demonstrate this mechanism in a delayed non-reciprocal Hopfield network, in which symmetric Hebbian couplings store $P = \alpha N$ memories while a delayed cyclic coupling drives their autonomous sequential replay. As the non-reciprocal mixture parameter λ increases, in the thermodynamic limit the model experiences a phase transition from static memory to dynamic sequence retrieval; we show that in the finite-size system this phase transition is actually decomposed into two transitions marking the end of static retrieval and the birth of a limit cycle. We show that those two transitions correspond to statistics of a quenched bifurcation ensemble. When λ increases, each memory fixed point continuously twists towards its successor and terminates at its own quenched saddle-node threshold $\lambda_c(\mu)$. Within one sample these P thresholds form a near-Gaussian, quasi-independent ensemble with a universal standardized shape: the load sets its center, the size sets its width, which narrows close to $N^{-1/2}$. The two macroscopic phase boundaries of the network are two quantiles of this single law: a typical threshold ends extensive static recall, while the sample maximum removes the last pinning barrier and releases sequential recall - we show indeed that the static extreme coincides with the dynamical onset in all disorder realizations we sampled. The terminal memories die on a directly closed invariant recall circle and the last saddle-node is a SNIC that depins a traveling orbit reached from all 100 stored patterns, while the delay re-aligns the retrieved states so that the orbit recalls every binary pattern exactly, with delay-paced period $T_1 = \tau + t_{\text{esc}}(\lambda)$, where $t_{\text{esc}}(\lambda)$ is the delay-independant effect of the final SNIC bifurcation on the limit cycle period. The window between the two transitions hosts a stationary hyperchaotic attractor coexisting with the few remaining retrieval states. Suppressing the delay, or correlating the patterns, destroys or relocates the barriers, isolating the two ingredients of the mechanism: a localized traveling front and discrete heterogeneous obstacles. The time delay can however be implicit, using a layered Markovian architecture showing an effective delay proportional to depth, thus giving birth to the sequential memory recall. As $N \rightarrow \infty$, the bifurcation ensemble variance goes to zero, the two boundaries merge and the selection mechanism becomes invisible, a concrete illustration of how a unique thermodynamic critical point can conceal the universal intra-sample statistics that built it.

Contents

1	Introduction: a sample is an ensemble	2
1.1	One microscopic distribution, two macroscopic transitions	2
1.2	The model: delayed non-reciprocal Hopfield networks	3
1.3	Exact dimensionality reduction to the pattern subspace	4
1.4	Global phase organization	5

2	Quenched ensemble of saddle-node bifurcations	6
2.1	Linear stability analysis of memory fixed points	6
2.2	One memory, one quenched threshold	8
2.3	Universal branch geometry and disordered memories	8
3	The threshold probability law and its order statistics	10
3.1	A universal position–scale family fixed by α and N	10
3.2	Two quantiles, two transitions	11
4	The extreme threshold is the dynamical onset	12
4.1	Static prediction versus dynamical onset, seed by seed	12
4.2	Saddle-node slowing on both sides of onset	13
4.3	A delay-paced relay clock	14
4.4	The cycle never destabilizes: Floquet analysis	14
5	One invariant recall circle	15
5.1	Direct closure of the terminal connection	15
5.2	All hundred memories reach the same orbit	16
5.3	Exact delay-induced sequential recall	16
6	Stationary hyperchaos between the two transitions	17
6.1	Coexistence with residual retrieval states	17
6.2	The hyperchaotic strange attractor	18
6.3	Stationarity: no aging on the resolved windows	19
7	What the mechanism requires: finite delay and discrete barriers	19
7.1	Short delays delocalize the recall front	19
7.2	A correlated Markov seam relocates the extreme barrier	21
7.3	Smooth periodic correlations remove pinning altogether	21
8	Implicit delay from Markovian depth, and general implications	21
8.1	A layered architecture with implicit memory	22
8.2	Depth sets the relay clock	23
9	Conclusion: from one neural network to a general finite-size mechanism	24

1 Introduction: a sample is an ensemble

1.1 One microscopic distribution, two macroscopic transitions

The statistical mechanics of disordered systems is usually organized around self-averaging: in the thermodynamic limit, a single large sample behaves like the disorder average, and a phase boundary is one number. At finite size this picture can fail in a structured way. When one realization contains many local states that are equivalent in the clean or averaged description, quenched fluctuations split their instabilities into a sample-specific ensemble of bifurcations: a set of P critical values coexisting inside a single system. Collective observables need not all respond to the same member of this set. An extensive observable follows its bulk; an observable that

requires every local state to have yielded follows its maximum. Distinct macroscopic transitions then emerge as distinct order statistics of one internal law, rather than as repeated estimates of one uncertain critical point. Here we resolve such a threshold ensemble bifurcation by bifurcation inside one disordered dynamical system, and follow how its quantiles become phase boundaries.

We demonstrate the mechanism in delayed non-reciprocal Hopfield networks: N -dimensional neural networks that store $P = \alpha N$ patterns through symmetric Hebbian couplings and replay them in a prescribed cyclic order through delayed non-reciprocal sequence-retrieving couplings. We study the non-vanishing load regime $\alpha > 0$, with typical value $\alpha = 0.05$ in this work. As the non-reciprocal mixture parameter λ grows, the static memory fixed points get twisted and lose linear stability each at its own pace, characterized by a pattern-dependent terminal threshold. Let $\lambda_c(\mu)$ denote the terminal threshold of the static retrieval state connected continuously to memory μ . Crossing the bulk of $\{\lambda_c(\mu)\}_{\mu=1}^P$ destroys an extensive fraction of static memories: this is the end of the static retrieval phase. Autonomous sequential recall, in contrast, is an all-or-nothing observable: a traveling cycle that must visit the full sequence remains pinned as long as a single obstruction survives. Its onset should therefore be controlled by the extreme threshold

$$\lambda_* = \max_{1 \leq \mu \leq P} \lambda_c(\mu). \quad (1)$$

At finite N , the typical threshold and λ_* are separated by an observable λ window, and the intermediate regime is a phase of its own. The measured narrowing of the threshold distribution is consistent with the two transitions merging as $N \rightarrow \infty$. Finite size is here not a correction to be scaled away; it is the magnifying glass that exposes the microscopic sequence of events from which the collective transition is born.

1.2 The model: delayed non-reciprocal Hopfield networks

We consider N continuous neurons $u_i(t)$ with firing rates $\phi_i(t) = \tanh[\beta u_i(t)]$, evolving under the delay differential equation (DDE) directly inspired from Hopfield's continuous-time evolution for classical symmetric hopfield networks:

$$t_0 \dot{\mathbf{u}}(t) = -\mathbf{u}(t) + (1 - \lambda)J\phi(t) + \lambda K\phi(t - \tau). \quad (2)$$

Here t_0 is the single-neuron relaxation time, τ the transmission delay, and $\lambda \in [0, 1]$ mixes energetic symmetric storage with delayed non-equilibrium dynamical recall. For unbiased binary patterns $\xi^\mu \in \{-1, +1\}^N$, with the cyclic convention $\xi^{P+1} = \xi^1$, the couplings are

$$J = \frac{1}{N} \sum_{\mu=1}^P \xi^\mu (\xi^\mu)^\top, \quad K = \frac{1}{N} \sum_{\mu=1}^P \xi^{\mu+1} (\xi^\mu)^\top. \quad (3)$$

The symmetric Hebbian matrix J makes each learned pattern a static retrieval state; the non-reciprocal matrix K maps activity aligned with ξ^μ toward its successor $\xi^{\mu+1}$. The delay is the essential third ingredient: it lets each memory be fully condensed before the sequential drive acts, a role made precise by the relay clock of Sec. 4.3 and by the short-delay experiments of Sec. 7.1. Non-reciprocity places the model outside gradient dynamics (there is no energy landscape to descend) connecting it to the broader physics of non-reciprocal phase transitions.

Figure 1 summarizes the two computational regimes. Below the loss of static recall, the network sits at a time-independent memory that already leans toward its successor; above the sequential onset, the same architecture supports a localized traveling wave visiting all memories

A Hopfield network that can also replay its memories in order

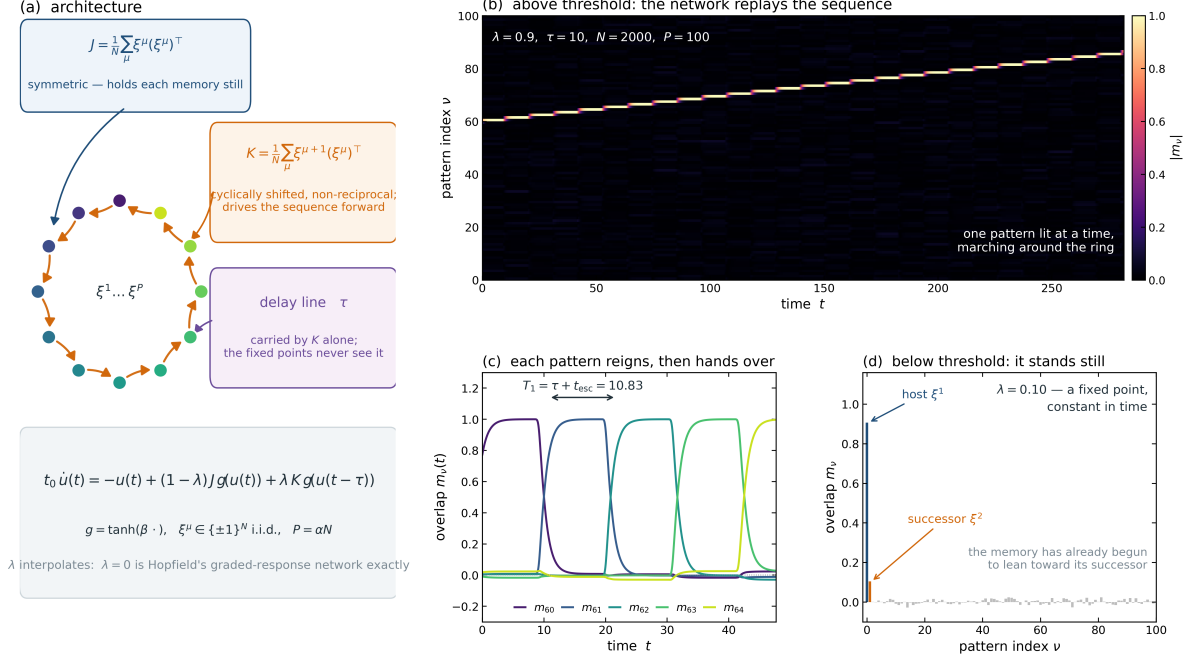


Figure 1: **A Hopfield network that can replay its memories in order.** (a) Symmetric Hebbian couplings (teal) stabilize each stored pattern, while the delayed cyclic coupling (red) points from each pattern to its successor. The delay acts only on the successor pathway, so it changes the dynamics without changing the fixed-point equation. (b) Above the sequential-recall threshold, a single localized activity peak traverses the complete memory ring. (c) Consecutive overlaps display the relay clock $T_1 = \tau + t_{\text{esc}}$: each memory remains expressed during one delay interval and then hands activity to its successor. (d) Below threshold, the same directed coupling tilts a retrieval state toward its successor without dislodging it.

in order. We monitor both regimes through the analog overlaps

$$m_{\mu}(t) = \frac{1}{N} (\xi^{\mu})^{\top} \tanh[\beta \mathbf{u}(t)] \quad (4)$$

and the binary readout overlaps

$$b_{\mu}(t) = \frac{1}{N} (\xi^{\mu})^{\top} \text{sgn}[\mathbf{u}(t)]. \quad (5)$$

Unless stated otherwise, branch-resolved numerical results refer to the reference realization

$$N = 2000, \quad P = 100, \quad \alpha = 0.05, \quad \beta = 20, \quad \tau = 10, \quad (6)$$

with disorder seed 42 and time in units of t_0 .

1.3 Exact dimensionality reduction to the pattern subspace

All quantitative results below rest on an exact, sample-specific dimensional reduction. Write the patterns as the rows of $X \in \{-1, +1\}^{P \times N}$ and let S be the cyclic shift, $S \mathbf{e}_{\mu} = \mathbf{e}_{\mu+1}$. We assume the Gram matrix $XX^{\top}$ is invertible, as holds with probability approaching one at the loads studied here; then $J = N^{-1} X^{\top} X$ and $K = N^{-1} X^{\top} S X$, and the decomposition $\mathbf{u} = X^{\top} \mathbf{a} + \mathbf{w}$ with $\mathbf{w} \perp \text{range}(X^{\top})$ is unique. Both local fields in Eq. (2) lie in $\text{range}(X^{\top})$, so the transverse

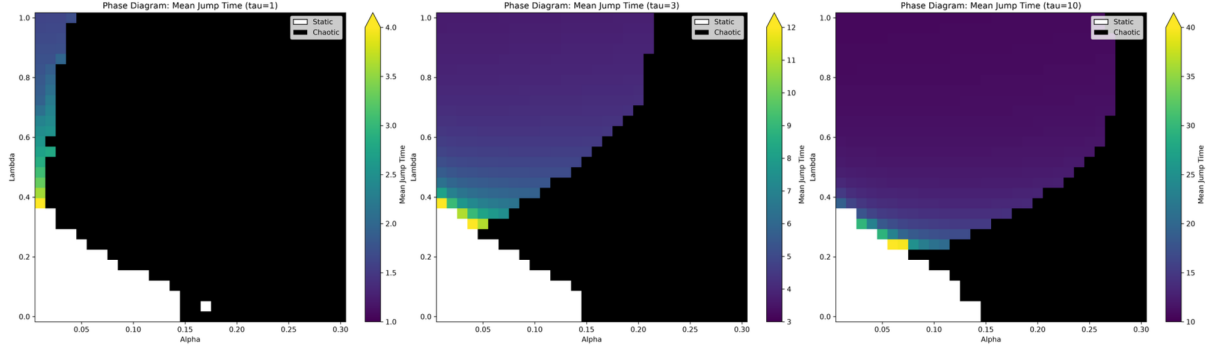


Figure 2: **Global static-to-sequential phase organization.** Operational phase diagrams in the (α, λ) plane for $\tau/t_0 = 1, 3, 10$ (left to right). White: persistent retrieval of the initial pattern. Color: ordered forward relays, shaded by mean relay duration. Black: failure of both criteria. The color scales show that the coherent relay time grows with the imposed delay. This survey organizes the regimes phenomenologically; all quantitative thresholds below come from the continuation machinery of Secs. 2–4.

component obeys exactly

$$t_0 \dot{\mathbf{w}} = -\mathbf{w}. \quad (7)$$

The realized pattern subspace is invariant and globally attracting. On it, the dynamics closes exactly on the P overlap coordinates:

$$t_0 \dot{\mathbf{a}}(t) = -\mathbf{a}(t) + (1 - \lambda) \mathbf{m}[\mathbf{a}(t)] + \lambda S \mathbf{m}[\mathbf{a}(t - \tau)], \quad (8)$$

with the sample-specific nonlinear overlap map

$$\mathbf{m}(\mathbf{a}) = \frac{1}{N} X \tanh(\beta X^\top \mathbf{a}). \quad (9)$$

Equation (8) is a finite-sample reduction, not a disorder-averaged closure: every cross-overlap of the realized patterns survives inside $\mathbf{m}$, and it is precisely this retained quenched interference that will split the bifurcations. On the stored certification trajectory, full and reduced solutions agree to 10^{-15} . Because the transverse dynamics is simply $t_0 \delta \dot{\mathbf{w}} = -\delta \mathbf{w}$, every nontrivial fixed-point, Floquet and Lyapunov spectrum of the N -dimensional DDE lives in the P -dimensional delayed system, which makes the exhaustive bifurcation characterization of this article feasible.

1.4 Global phase organization

Before resolving microscopic boundaries, we locate the main dynamical regimes. Starting from one stored memory, the (α, λ) plane was scanned at $\tau/t_0 \in \{1, 3, 10\}$ (Fig. 2). White points remain in static recall; colored points advance through the prescribed sequence and are shaded by their mean relay time; black points meet neither criterion and correspond to a paramagnetic chaotic regime. Increasing the delay opens a broad coherent sequential-recall region whose relay time grows in proportion to τ ; at $\tau = 1$ the ordered region is strongly eroded and irregular dynamics dominates, as the continuous-time wave traveling the sequence is unable to remain condensed on a single pattern.

The survey used $N = 2500$, $\beta = 20$, a 30×30 grid over $\alpha \in [0.01, 0.30]$ and $\lambda \in [0, 1]$, Euler integration with $\Delta t = 0.05$, and $J_{ii} = 0$. It is a phenomenological map, which this work explains through the stability analysis of the local dynamics. All continuations and critical values reported

From quenched memory folds to the global recall cycle

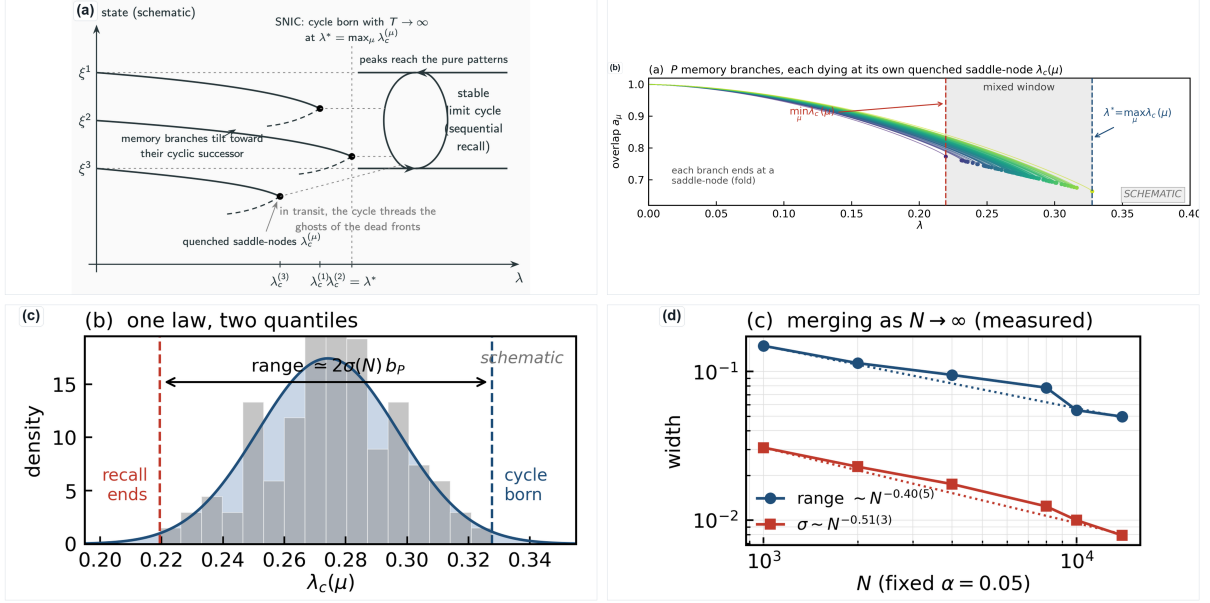


Figure 3: **From quenched local folds to a global recall cycle.** (a) Schematic organization: memory fixed-point branches disappear at distinct thresholds $\lambda_c(\mu)$, while the moving orbit threads their ghosts and is released at the terminal barrier. (b) The $P = 100$ primary branches of the reference network, represented by their self-overlap, terminate across a finite threshold interval; dashed lines mark its edges. (c) The same ensemble as a distribution: a typical quantile ends extensive static recall, the upper extreme predicts cycle onset. (d) At fixed $\alpha = 0.05$, both the full range and the standard deviation shrink over $N = 10^3 - 1.4 \times 10^4$; dotted lines are power-law fits. Panels (b,c): $(N, P, \beta, \tau/t_0) = (2000, 100, 20, 10)$, seed 42.

below use the low-rank couplings of Eq. (3), double precision, and higher-order integration.

The apparently smooth white-to-color boundary of Fig. 2 conceals the finite-size structure that is the subject of this article. We will show that as λ grows, memory branches disappear one at a time; the bulk of their thresholds ends extensive static recall, the largest one releases the limit cycle, and residual attractors and chaotic dynamics fill the interval in between. We now resolve this ensemble of individual bifurcations.

2 Quenched ensemble of saddle-node bifurcations

The static retrieval states of one network do not die together. Figures 3 and 4 establish the elementary fact on which everything else is built: the primary branch connected to each stored pattern ends at its own quenched saddle-node threshold, and the P thresholds of one sample span a wide, structured interval.

2.1 Linear stability analysis of memory fixed points

Fixed points do not see the delay. At a fixed point, delayed and instantaneous activities coincide, $\mathbf{u}(t - \tau) = \mathbf{u}(t) = \mathbf{u}^*$, so the delay drops out of the equilibrium equation:

$$\begin{aligned} \mathbf{F}(\mathbf{u}; \lambda) &= -\mathbf{u} + C(\lambda) \tanh(\beta \mathbf{u}) = \mathbf{0}, \\ C(\lambda) &= (1 - \lambda)J + \lambda K. \end{aligned} \tag{10}$$

The entire fixed-point landscape (every equilibrium and every fold location) is therefore delay-independent; only stability margins and the surrounding dynamics depend on τ .

Instantaneous versus delayed linearization. Linearizing the delay-free part of the flow around $\mathbf{u}^*$, with the diagonal gain matrix $D = \beta \text{diag}\{\text{sech}^2(\beta u_i^*)\}$ (the derivative of $\tanh \beta u$ at the fixed point), gives the ordinary Jacobian

$$M = -I + C(\lambda)D. \quad (11)$$

For the delayed system, the correct object follows from linearizing the full DDE (2): a perturbation $\delta \mathbf{u}(t)$ obeys

$$t_0 \delta \dot{\mathbf{u}}(t) = -\delta \mathbf{u}(t) + (1 - \lambda)JD \delta \mathbf{u}(t) + \lambda KD \delta \mathbf{u}(t - \tau). \quad (12)$$

Inserting the exponential ansatz $\delta \mathbf{u}(t) = e^{zt} \mathbf{v}$ (the delayed term contributes $e^{z(t-\tau)} \mathbf{v} = e^{-z\tau} e^{zt} \mathbf{v}$) turns Eq. (12) into the algebraic condition $\Delta(z) \mathbf{v} = \mathbf{0}$ with the characteristic matrix

$$\Delta(z) = (t_0 z + 1)I - (1 - \lambda)JD - \lambda e^{-z\tau} KD. \quad (13)$$

Because of the transcendental factor $e^{-z\tau}$, the equation $\det \Delta(z) = 0$ has countably infinitely many roots (with real parts bounded above and accumulating at $\text{Re } z \rightarrow -\infty$); the fixed point is linearly stable if and only if every root satisfies $\text{Re } z < 0$.

Exact reduction of the characteristic problem. The infinite root hunt would be hopeless at $N \times N$ scale, but the couplings have rank at most P . Writing $JD = N^{-1} X^\top X D$ and $KD = N^{-1} X^\top S X D$, the two z -dependent terms of $\Delta(z)$ combine into the single low-rank product

$$(1 - \lambda)JD + \lambda e^{-z\tau} KD = \frac{1}{N} X^\top W(z) X D, \quad W(z) = (1 - \lambda)I_P + \lambda e^{-z\tau} S. \quad (14)$$

Sylvester's determinant identity, $\det(cI_N + UV) = c^{N-P} \det(cI_P + VU)$ for $U \in \mathbb{R}^{N \times P}$ and $V \in \mathbb{R}^{P \times N}$, applied with $U = X^\top W(z)$ and $V = XD/N$, collapses the transcendental $N \times N$ problem to $P \times P$:

$$\det \Delta(z) = (t_0 z + 1)^{N-P} \det T_P(z), \quad (15)$$

$$T_P(z) = (t_0 z + 1)I_P - [(1 - \lambda)I_P + \lambda e^{-z\tau} S] \frac{XDX^\top}{N}. \quad (16)$$

The prefactor $(t_0 z + 1)^{N-P}$ is the trivial bulk of transverse roots $z = -1/t_0$ (the linearized version of the exact transverse decay Eq. (7)) while all nontrivial characteristic roots solve $\det T_P(z) = 0$. They are thus accessible without ever forming an $N \times N$ spectrum.

The fold discriminator. Evaluating Eq. (13) at $z = 0$ (where $e^{-z\tau} = 1$) gives the exact identity

$$\Delta(0) = I - CD = -M : \quad (17)$$

a zero root of the delay equation is exactly a zero eigenvalue of the static continuation problem. A fold can therefore be located on the equilibrium branch and certified independently of the delay spectrum, and its position is automatically τ -independent. Conversely, the only way the delay could change the death of a branch is a complex pair crossing the axis before the fold - a delay-induced Hopf bifurcation - which must be checked on the delayed spectrum of Eq. (16).

That doesn't happen in the regime studied in this paper: over a grid $\alpha \in \{0.01, 0.03, 0.05, 0.07, 0.10\}$, $\tau/t_0 \in \{0, 0.5, 1, 2, 4, 6, 8, 10, 12, 15\}$, every terminal point carries a real zero mode. The $\alpha = 0.10$ edge of the grid deserves a caveat. It lies close to the retrieval-capacity boundary ($\alpha_c \approx 0.14$), where the single-realization fold is no longer a well-defined quantity: the fitted fold location drifts strongly with N (from ≈ 0.18 at $N = 10^3$ to ≈ 0.05 at $N = 10^4$), the seed-to-seed scatter reaches $\sim 60\%$ of the mean, and some realizations produce spurious unstable starting states - the fingerprint of the rough many-state landscape near capacity. The $\alpha = 0.10$ row of the grid is therefore reported as indicative of the qualitative verdict (real zero mode, no Hopf bifurcation) rather than as a quantitative threshold; near-capacity physics requires analytic treatment. Away from that edge, in this low-but-nonnegligible-load regime, memory branches terminate in folds with no resolved Hopf bifurcation preceding them. The Hopf margin (distance of the rightmost complex pair to the imaginary axis at the fold) decreases with the delay roughly as $1/\tau$ but remains clearly positive at least for all $\tau \lesssim 10^2$ at $\alpha \leq 0.05$.

2.2 One memory, one quenched threshold

Newton continuation from each stored pattern at $\lambda = 0$ tracks its primary stable retrieval branch, together with the adjacent index-one saddle, up to the point where the smallest singular value of M vanishes. Carrying out this program for all $P = 100$ primary branches of the reference sample yields thresholds spanning a substantial interval:

$$\begin{aligned} \min_{\mu} \lambda_c(\mu) &= 0.2195, \\ \text{median}_{\mu} \lambda_c(\mu) &\simeq 0.2742, \\ \max_{\mu} \lambda_c(\mu) &= 0.3276, \end{aligned} \tag{18}$$

each localized by bisection to $\pm 2 \times 10^{-4}$, so the width of the interval, 0.11, is some fifty times larger than the uncertainty on any single threshold. The minimum belongs to pattern 28, the maximum to pattern 91; the branch of pattern 1 dies at $\lambda_c(1) = 0.2822$. These values coexist at fixed disorder realization: their spread is not sample-to-sample uncertainty about a unique critical point, it is an internal random object of a single network, and the central object of study of this work.

2.3 Universal branch geometry and disordered memories

How does a memory approach its death? Not by an abrupt saddle-node collision, but through a continuous tilting process. As λ grows, the sequence-retrieving drive pushes each retrieval state ξ^μ toward its successor $\xi^{\mu+1}$, long before the fold. Defining $\lambda_{50}(\mu)$ as the point where the successor overlap reaches half its terminal value, the 100 primary branches give

$$\text{median}_{\mu} \frac{\lambda_{50}(\mu)}{\lambda_c(\mu)} = 0.64, \quad \text{IQR} = [0.61, 0.66]. \tag{19}$$

The deformation is a broad precursor occupying most of the branch.

When λ is rescaled by each branch's own threshold and the tilt is normalized by its terminal value, the 100 trajectories collapse onto a quasi-single shape: disorder decides when a branch dies, but the normalized path to death is universal. The terminal tilt ratio ranges from 0.29 to 0.49 across the 100 branches, and correlates strongly with the threshold itself, $\text{corr}(r_c, \lambda_c(\mu)) = +0.75$ at $N = 10^4$ (+0.81 at $N = 2000$) (branches that die later die more tilted). At fixed λ_c , the residual scatter of r_c is almost entirely explained by the local quenched bond overlap $q_\mu = N^{-1} \xi^\mu \cdot \xi^{\mu+1}$:

Static folds reshape memories before they disappear

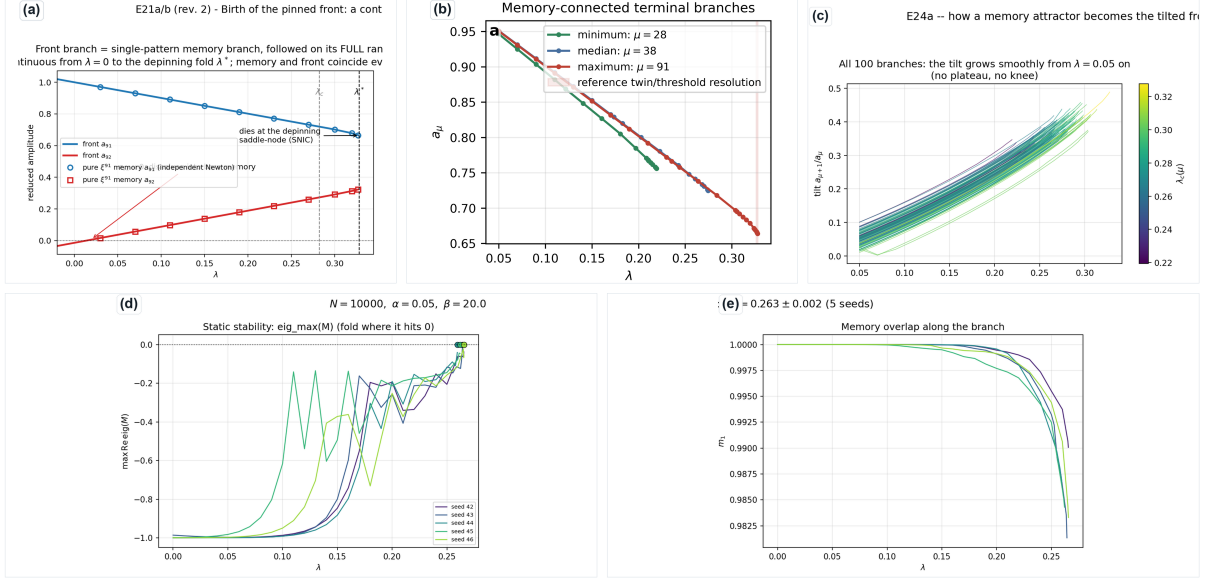


Figure 4: **Static folds reshape memories before they disappear.** (a) Along the terminal branch (memory 91), the self-amplitude decreases while the successor amplitude grows continuously up to the fold. (b) The branches with minimum, median and maximum λ_c show the same loss of self-amplitude. (c) For all 100 branches, the successor-to-current amplitude ratio rises smoothly; color encodes $\lambda_c(\mu)$. (d) At $N = 10^4$, the leading static eigenvalue reaches zero in five disorder samples. (e) Their memory overlaps stay close to unity before bending sharply near the fold: the saturated overlap readout hides the amplitude tilt of panels (a–c) until the fold itself (see text). Panels (a–c): reference network; (d,e): $\alpha = 0.05$, $\beta = 20$.

the partial correlation of (q_μ, r_c) at fixed λ_c is $+0.90$ at $N = 10^4$ ($+0.95$ at $N = 2000$). The pair $(\lambda_c(\mu), q_\mu)$ thus determines the death shape almost deterministically.

A related saturation effect explains why the overlap readout of Fig. 4(e) stays close to unity while the amplitudes tilt so strongly. The overlap $m_\mu = N^{-1} \xi^\mu \cdot \tanh(\beta \mathbf{u})$ is a saturated function of the state: at $\beta = 20$, $\tanh(\beta u_i)$ is within 10^{-4} of $\text{sgn}(u_i)$ whenever $|u_i| \gtrsim 0.5$. On a tilted branch, $\mathbf{u} \approx a_\mu \xi^\mu + a_{\mu+1} \xi^{\mu+1}$ with $a_\mu \approx 0.75$ and $a_{\mu+1} \approx 0.25$ near the fold, so $|u_i| \geq a_\mu - a_{\mu+1} \approx 0.5$ on every site and $\text{sgn}(u_i) = \xi_i^\mu$ everywhere: the sign structure, hence the saturated overlap, remains that of the pure memory until the successor amplitude approaches parity, which happens only in the immediate vicinity of the fold. The twisting is thus large and continuous in the subspace coordinates $\mathbf{a}$ (up to $r_c \approx 0.3$ – 0.5) while remaining almost invisible to the overlap until the terminal square-root collapse: both panels of Fig. 4 describe the same smooth distortion, read through two differently saturating observables.

Near the fold, the geometry is the generic saddle-node one. A local branch coordinate obeys the normal form $\dot{q} \sim A(\lambda - \lambda_c) + Bq^2$, and the saddle-node distance and the rightmost eigenvalue scale as $|\lambda - \lambda_c|^{1/2}$; direct continuation at $N = 10^4$ confirms the square-root collision, with fits on representative branches giving $R^2 = 0.93$ – 0.98 .

To avoid assigning a memory branch the label of its successor, memory labels are not read off the state at the fold, where the retrieved memory has become a distorted object; each label is inherited continuously from the $\lambda = 0$ retrieval state and checked through the overlap vector. This makes $\{\lambda_c(\mu)\}_{\mu=1}^P$ a well-defined quenched observable.

The picture emerging from this section is a competition between the symmetric and non-reciprocal couplings: while the symmetric term builds metastable retrieval states, the sequential

Thresholds form a finite-size random ensemble

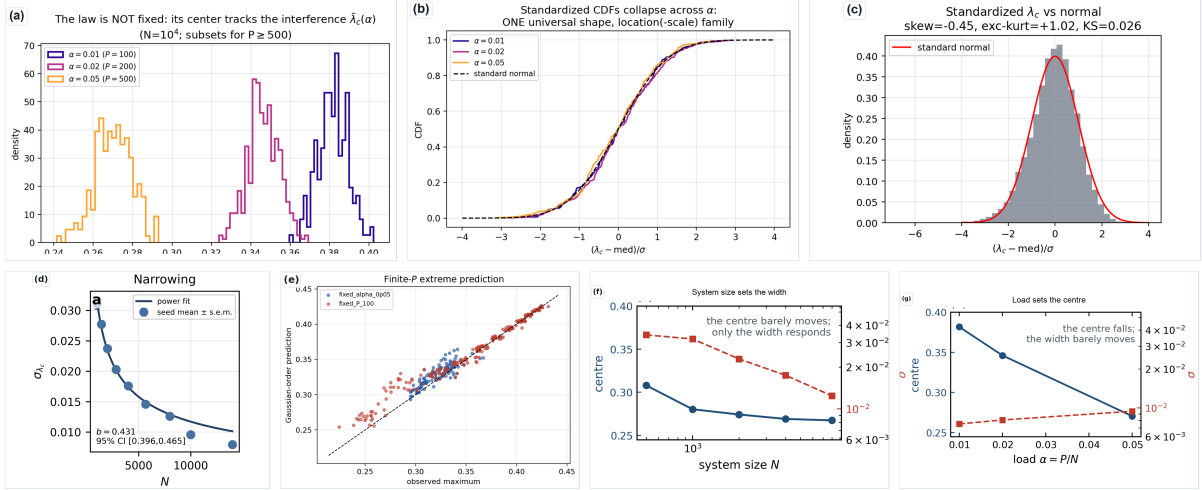


Figure 5: **The finite-size law of memory thresholds.** (a) At $N = 10^4$, increasing α shifts the unstandardized distribution. (b) After centring and scaling, the empirical CDFs collapse onto a common near-Gaussian shape; (c) pooled standardized density with normal reference. (d) A seed-aware fit of the width gives exponent 0.43, 95% CI $[0.40, 0.47]$: a CLT-like narrowing. (e) Observed sample maxima versus the finite- P Gaussian extreme-value prediction, Eq. (24). (f) Varying N mainly changes the width; (g) varying α mainly changes the center. Unless stated in a panel, $\beta = 20$; uncertainties are resampled over disorder seeds, not over patterns within one seed.

term deforms each of them along a universal path toward its successor, as finite-size quenched cross-talk assigns each branch its private saddle-node threshold. The dynamical problem has become a statistical one: are those quenched threshold drawn iid from a marginal probability distribution, if so which one, and which of its statistics governs each collective phase transition?

3 The threshold probability law and its order statistics

The P thresholds of one finite-size network form a quenched random field on the memory ring, not an imposed collection of independent variables. Their one-point law nevertheless turns out to be remarkably simple, as summarized in Fig. 5: the load sets its center, the system size sets its width, and its standardized shape is universal across realizations and across loads. The thresholds behave as near-independent draws from a single near-Gaussian law whose width narrows with N , so that the sample maximum is accurately predicted by Gaussian extreme-value theory. This section establishes the law; the next sections show that the network’s two phase boundaries read two different quantiles of it.

3.1 A universal position–scale family fixed by α and N

For each realization, let $\bar{\lambda}_c$ and σ_{λ_c} be the mean and standard deviation over its memory branches, and standardize

$$z_\mu = \frac{\lambda_c(\mu) - \bar{\lambda}_c}{\sigma_{\lambda_c}}. \quad (20)$$

At $N = 10^4$, raising the load from $\alpha = 0.01$ to 0.02 and 0.05 moves the median threshold from 0.38 to 0.35 and 0.27—tracking the mean interference between patterns—while the widths barely respond: $\sigma_{\lambda_c} = 0.0076, 0.0081, 0.0094$ (a 24% increase for a five-fold increase of α , to be

compared with the factor $\simeq 2.4$ separating σ_{λ_c} at $N = 2000$ from $N = 10^4$ at fixed α). Meanwhile the standardized laws are statistically indistinguishable: two-sample Kolmogorov–Smirnov tests comparing the standardized threshold samples of the three loads pairwise (0.01 vs 0.02, 0.01 vs 0.05, 0.02 vs 0.05) accept the hypothesis of a common distribution with $p = 0.85, 0.87$ and 0.25 . Within the tested range, the ensemble is therefore a position-scale family: a single fixed standardized distribution, shifted by a load-dependent location and stretched by a size-dependent scale,

$$\lambda_c(\mu) \approx m(\alpha) + \sigma(N, \alpha) Z_\mu, \quad (21)$$

where the standardized variable Z_μ has a load-independent, near-Gaussian law. Pooling the standardized fixed-load data for $N \geq 4000$ (8764 branch thresholds) gives skewness -0.45 , excess kurtosis 1.0 , and a maximum CDF distance 0.026 from a fitted normal. The Gaussian captures the bulk and the leading extreme behavior, despite the small negative skew and excess tails.

This empirical structure has a direct physical reading. Each threshold is a smooth functional of the $\sim \alpha N$ quenched cross-overlaps $N^{-1}(\xi^\mu)^\top \xi^\nu$ felt by its branch, each of variance $1/N$; a sum of many weakly dependent contributions gives a central-limit, near-Gaussian shape. The same reasoning suggests near-independence around the ring, which is verified directly: the lag-one autocorrelation of neighboring thresholds is -0.009 at $N = 10^4$, so the ensemble is quasi-i.i.d. As for the center depending mainly on the load, this is the finite-size shadow of the thermodynamic limit: as the width shrinks to zero all thresholds collapse onto the single critical value $\lambda_c^\infty(\alpha)$ of the infinite system, which the macroscopic phase diagrams show to depend strongly on α .

The width narrows with size as a power law,

$$\sigma_{\lambda_c}(N) \propto N^{-b}, \quad b = 0.43, \quad 95\% \text{ CI } [0.40, 0.47] \text{ at } \alpha = 0.05, \quad (22)$$

where the exponent and its confidence interval are seed-aware: the fit treats each disorder realization as a single statistical unit and the uncertainty is obtained by bootstrap resampling of whole realizations, never by pooling the P correlated thresholds of one sample as if they were independent measurements. The interval sits below the CLT value $1/2$, which we read as a finite-size and within-sample-dependence correction to a perfect central-limit scaling.

The max–min range of the sample narrows too, more slowly: it decreases from 0.15 at $N = 1000$ to $0.078, 0.055$ and 0.050 at $N = 8000, 10^4$ and 1.4×10^4 . For P near-independent Gaussian draws, extreme-value theory predicts $\text{range}(N) \simeq 2 \sigma(N) a_P$ with the slowly growing factor $a_P \sim \sqrt{2 \ln P}$, $P = \alpha N$ —a power law times a logarithmic correction. The data agree with this form: fitting $\text{range}(N)/\sqrt{2 \ln(\alpha N)} \propto N^{-b_{\text{gap}}}$ over the scaling campaign gives $b_{\text{gap}} = 0.42$, seed-aware 95% CI $[0.37, 0.47]$, statistically indistinguishable from the width exponent of Eq. (22). The range is thus fully accounted for by the one-point width and Gaussian order statistics.

3.2 Two quantiles, two transitions

The law of Eq. (21) produces two physically distinct scales. The fraction of primary branches (those connected continuously to the $\lambda = 0$ memories) surviving at a given λ is the complementary CDF of the threshold law,

$$f_{\text{stat}}(\lambda) = \frac{1}{P} \sum_{\mu=1}^P \Theta[\lambda_c(\mu) - \lambda], \quad (23)$$

and falls as λ crosses the bulk. This crossover, set by a typical quantile such as the median, is the end of the extensive static-recall phase, even though individual fixed points keep disappearing on both sides of it.

Sequential recall obeys a stricter logic. A wave that must visit all P memories is held by the last surviving barrier, so its onset is predicted, to leading order, by the sample maximum λ_* of Eq. (1). For quasi-i.i.d. Gaussian thresholds the finite- P extreme-value estimate is

$$\begin{aligned}\mathbb{E}[\lambda_*] &\simeq \bar{\lambda}_c + \sigma_{\lambda_c} a_P, \\ a_P &= \sqrt{2 \ln P} - \frac{\ln \ln P + \ln(4\pi)}{2\sqrt{2 \ln P}}.\end{aligned}\tag{24}$$

Across the scaling data this tracks the observed maxima with mean signed error +0.005 and maximum absolute error 0.024 [Fig. 5(e)] - to be compared with the sample-to-sample spread of the maxima themselves, of order 0.01–0.03: the parameter-free Gaussian prediction is accurate at the level of the natural fluctuations, with a slight systematic over-prediction consistent with the small negative skew.

At finite N , the typical and maximal quantiles therefore bracket a dynamical window in which extensive static recall has ended while the last quenched barrier still pins the ordered orbit. Because $P = \alpha N$ grows only linearly while σ_{λ_c} shrinks approximately as $N^{-1/2}$,

$$\lambda_* - \bar{\lambda}_c \sim \sigma_{\lambda_c} \sqrt{2 \ln(\alpha N)} \longrightarrow 0 : \tag{25}$$

The two transitions merge in the thermodynamic limit, leaving a single critical point that conceals the selection mechanism.

4 The extreme threshold is the dynamical onset

Order statistics identify λ_* as the leading onset predictor; they do not by themselves establish which bifurcation releases the orbit. We therefore confront two fully independent measurements in each disorder realization. Static continuation finds the largest memory threshold without any reference to a recall trajectory; a descending dynamical scan, initialized on the moving branch, locates the last ordered cycle, the first stopping state, and the slowest relay. Figure 6 shows that the two procedures select the same quenched memory.

4.1 Static prediction versus dynamical onset, seed by seed

The identity test was run on 20 independent networks with $N = 2000$, $P = 100$, $\beta = 20$, $\tau = 10$. In all of them, the independently selected static maximizer, the slowest relay bond, and the final arrest memory are the same pattern. One realization has overlapping leading static brackets, so the identity of its maximum is not resolvable; the remaining 19 realizations satisfy

$$\left| \lambda_*^{\text{static}} - \lambda_{\text{onset}}^{\text{dyn}} \right| < 10^{-3}, \tag{26}$$

with maximum retained discrepancy 7.3×10^{-4} —while the thresholds themselves are spread over an interval of order 10^{-1} : the agreement is two orders of magnitude finer than the scale of the quantity predicted.

In the reference sample the maximizer is memory 91 and $\lambda_* = 0.328$. Below this value a moving initial history ultimately stops at that memory; above it the trajectory passes and completes the ring. The other 99 thresholds shape the transient sequence and the roster of coexisting fixed points, but no other primary branch sets the onset of global motion in this realization. The maximum of a statistical ensemble thus predicts a dynamical phase boundary.

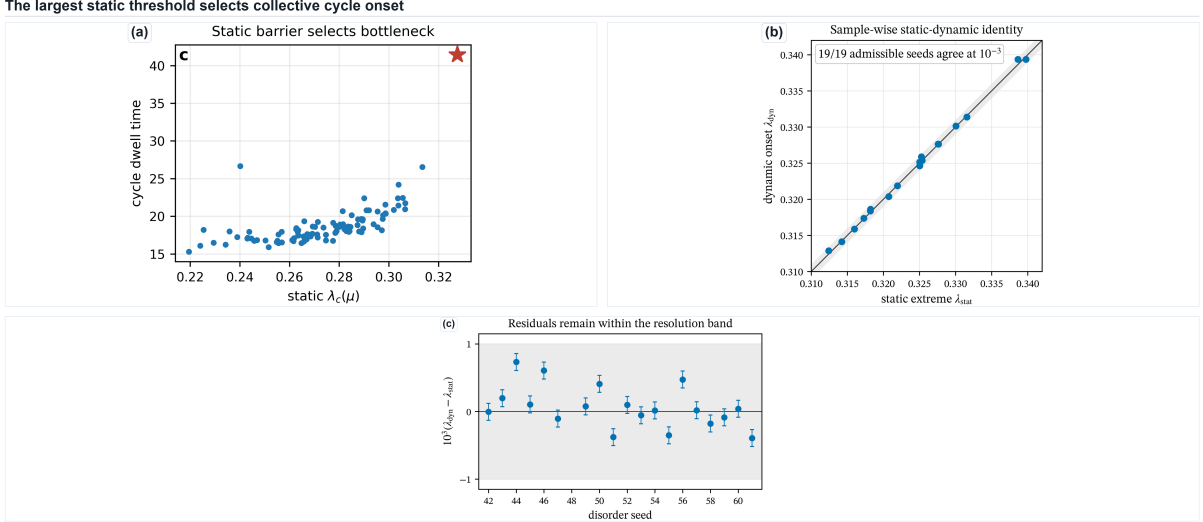


Figure 6: **Static extreme selection of the dynamical onset.** (a) Reference network: relay dwell time versus the independently continued $\lambda_c(\mu)$; the largest threshold (star, memory 91) selects the bottleneck. (b) Across 19 independently admissible disorder realizations, the operational dynamical onset lies on the static-extreme diagonal at resolution 10^{-3} (grey band). (c) Seed-resolved residuals and onset-bracket uncertainties all remain inside the band; the pattern identity test succeeds in all 20 tested networks. $(N, P, \beta, \tau) = (2000, 100, 20, 10)$.

4.2 Saddle-node slowing on both sides of onset

If the onset is a saddle-node collision on the orbit's path, its normal form dictates a passage time diverging as $|\lambda - \lambda_*|^{-1/2}$ from above and a soft real eigenvalue vanishing as $(\lambda_* - \lambda)^{1/2}$ from below. Both signatures are observed, and the slowing down happens only at the extreme bond.

From above, the critical 91-to-92 relay slows dramatically while the rest of the ring barely notices: for $\lambda = 0.332, 0.330, 0.329, 0.328$, the median relay times are 17.8, 18.0, 18.1, 18.3, while the critical relay takes 35.6, 41.5, 46.9, 72.9. A direct ghost fit gives

$$t_{91 \rightarrow 92} \simeq \frac{C}{\sqrt{\lambda - \lambda_g}}, \quad C = 2.3 \pm 0.2 \text{ (a 9\% relative uncertainty)}, \quad \lambda_g = 0.3262, \quad (27)$$

with λ_g consistent with the independently bracketed onset $\lambda \in [0.327, 0.328]$ to within twice that scan's bracket width. This is the extreme-value structure at work dynamically: one bond dies, ninety-nine stay passable, and only the maximum is visible in the divergence.

A logarithmic divergence (which would signal a competing homoclinic scenario) is rejected by model comparison, and the multi-seed data support saddle-node square-root slowing with finite-window corrections: the free exponent is between 0.43 and 0.51 at 95% confidence, an interval that contains $1/2$ and excludes both 0 (logarithmic) and 1 (linear).

From below, the leading real eigenvalue of the pinned terminal state softens as

$$\zeta_{\max} \simeq -2.1\sqrt{\lambda_* - \lambda}, \quad \lambda_*^{\text{spectral}} \simeq 0.328, \quad (28)$$

reaching zero at the same fold that releases the orbit. A caution applies to the smallest distances shown in Fig. 7(c): points with $\delta = \lambda_* - \lambda \lesssim 10^{-4}$ lie below the resolution at which λ_* itself is defined (bisection brackets $\pm 2 \times 10^{-4}$), so the quantitative support for the square-root law is the

The collective transition is a saddle-node on an invariant circle

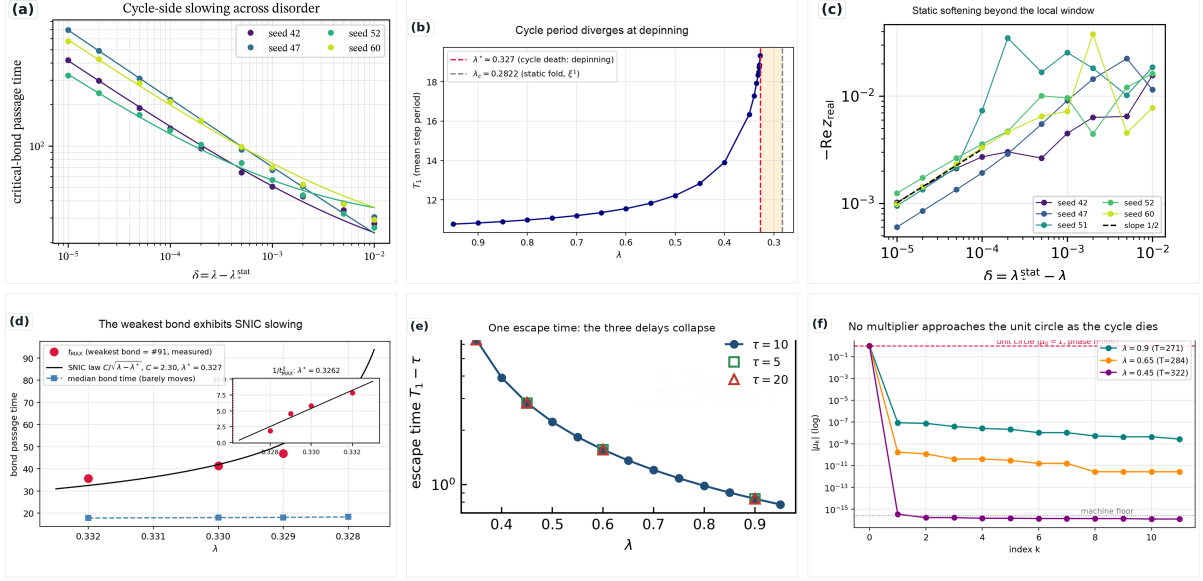


Figure 7: **The terminal bifurcation is a saddle-node on an invariant circle.** (a) Cycle-side passage times for four admissible seeds versus distance from the independently continued extreme. (b) The mean relay period separates the disappearance of a representative static fold from the depinning threshold. (c) The leading real eigenvalue softens with square-root slope on the static side; points below $\delta \simeq 10^{-4}$ are indicative only (see text). (d) Only bond 91 develops a divergent dwell time; the inset linearizes the $1/2$ law. (e) The escape time $T_1 - \tau$ extracted at $\tau = 5, 10, 20$ collapses onto a single curve: the delay enters the relay period additively and the escape time not at all. (f) Floquet spectrum of the sequential cycle: the phase multiplier stays on the unit circle while every nontrivial multiplier collapses by orders of magnitude on approach to λ_* , and none approaches $+1$ (Sec. 4.4). Panels (b), (d), (e) and (f): seed 42; (a) and (c) pool four and five disorder realizations respectively. $(N, P, \beta) = (2000, 100, 20)$.

decade $\delta \in [10^{-3}, 10^{-2}]$ together with the direct continuation fit of Eq. (28); the deeper points are shown as a consistency illustration only. The divergent passage above and the soft real mode below meet at one static–dynamic threshold.

4.3 A delay-paced relay clock

Away from the terminal divergence, every link of the sequential cycle has two temporally distinct stages: the successor signal cannot act before the delayed activity arrives, and after arrival the state needs a nonlinear local escape time, which diverges at the SNIC. The measured inter-memory period is their sum,

$$T_1(\lambda, \tau) = \tau + t_{\text{esc}}(\lambda). \quad (29)$$

The decomposition is exact to the resolution of the data: for $\tau = 5, 10, 20$, the extracted escape times collapse to three significant digits [Fig. 7(e)]. The delay sets the clock; the quenched terminal fold supplies the divergent local escape. In the dynamical phase the network acts as a pacemaker whose rhythm is architectural and whose bottleneck is statistical.

4.4 The cycle never destabilizes: Floquet analysis

Square-root slowing identifies a saddle-node ghost, but a fold of cycles could mimic the disappearance of the orbit. The Floquet spectrum of the cycle excludes this. Any local bifurcation of a limit

The recall orbit persists as a single invariant circle

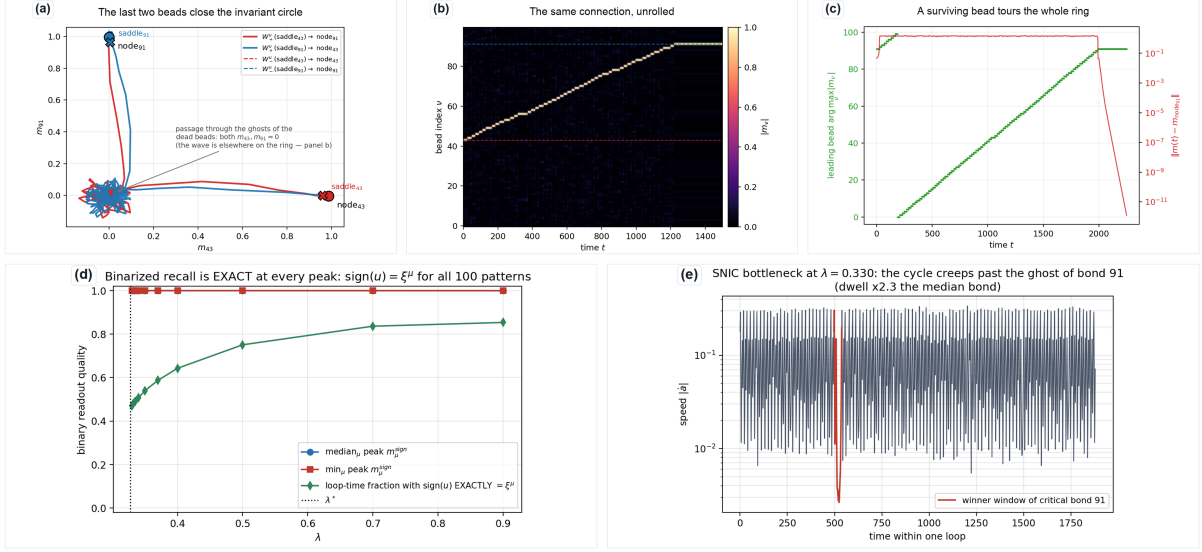


Figure 8: **The terminal invariant recall circle of the reference network.** (a) At $\lambda = 0.308$, the two forward unstable branches connect $43 \rightarrow 91$ and $91 \rightarrow 43$ in the (m_{43}, m_{91}) projection. (b) Unrolled over pattern index, the same connection is a single wave traversing the ring through the ghosts of the 98 vanished memories. (c) When only memory 91 survives, its forward branch tours the ring and returns to the node to machine precision. (d) On the released cycle, every pattern reaches binary overlap one at its own peak; the green curve is the fraction of a tour spent in an exact binary memory. (e) At $\lambda = 0.330$, the velocity minimum is concentrated at the ghost of bond 91. $(N, P, \beta, \tau) = (2000, 100, 20, 10)$, seed 42.

cycle announces itself by a nontrivial multiplier approaching the unit circle; here the multipliers do the opposite. A broad scan from $\lambda = 0.35$ to 0.95 finds a strongly stable cycle with the neutral phase multiplier retained, and near-threshold controls extend the conclusion to $\lambda = 0.335$ and 0.329 [Fig. 7(f)]. The dominant nontrivial contraction per tour $A = -\ln |\mu_1|$ actually grows, from 220 at $\lambda = 0.400$ to 420 at 0.335 and 440 at 0.329 , while the contraction rate per unit time stays finite (between about -0.16 and $-0.23 t_0^{-1}$): the multiplier collapses toward zero because the tour time diverges, not because the transverse dynamics weakens. No multiplier approaches $+1$. The cycle dies at full stability, by losing its route rather than its attractivity - the defining Floquet signature of a saddle-node on an invariant circle (SNIC), in which the equilibrium pair annihilates on the orbit. Establishing that the collision indeed lies on a closed invariant route is a global statement, which we now verify by direct construction.

5 One invariant recall circle

The SNIC diagnosis requires more than matching critical values: the terminal node and saddle must lie on a closed one-dimensional invariant connection. We construct that connection explicitly in the reference realization, and determine whether the released motion is unique.

5.1 Direct closure of the terminal connection

Just below λ_* , let $\mathbf{u}_s$ be the index-one saddle paired with the terminal memory node and $\mathbf{v}_u$ its unstable eigenvector. Launching trajectories at $\mathbf{u}_s \pm \epsilon \mathbf{v}_u$ and integrating the full reduced dynamics, one branch falls directly onto the nearby node. The other, remarkably, advances through the

entire ordered memory sequence (including the locations of all 98 vanished branches) and returns to the very node it left. At $\lambda = 0.323$ ($\lambda_* - \lambda = 0.005$) the tour takes $1991 t_0$ and closes with Gram-metric distance 3.5×10^{-14} ; at $\lambda = 0.318$ it takes $2138 t_0$ and closes within about 10^{-13} . For comparison, the state itself is an order-one vector, so these closure distances are at the level of accumulated round-off: the union of node, saddle and connecting branches is a numerically closed invariant circle.

One step deeper in the removal sequence, the construction repeats. At $\lambda = 0.3076$ two stable memory beads survive, patterns 43 and 91, and both forward links are certified: the unstable branch at 43 reaches node 91, that at 91 reaches node 43. The two-bead necklace is again a single closed circle threading the ghosts of the other 98 memories. These direct closures establish the cyclic organization at the terminal and penultimate stages of the reference realization. Farther below threshold, most local links are certified individually, but the complete many-bead object has not been closed end to end; the circle is thus established as the terminal organizing structure, and strongly supports the picture in which all memory states die on this common route.

The equilibrium branches close the local part of the argument: the stable node persists up to $\lambda = 0.328$ with self-overlap decreasing from 0.999 to 0.982, and the paired index-one saddle reaches the same parameter with overlap 0.982. Saddle-node coalescence at λ_* , the closed connection below it, and the cycle above it together place the last fold on the invariant circle: a SNIC.

5.2 All hundred memories reach the same orbit

Above λ_* the invariant route could in principle fragment into several coexisting cycles, each recruited by a subset of memories. It does not. Initializing the reduced dynamics independently at each of the 100 stored patterns, every trajectory becomes periodic, visits all 100 memories with ring step +1, and identifies memory 91 as the slowest relay. Comparing orbits rather than phases - each trajectory aligned at the refined peak of memory 0 and resampled on a common phase grid - the waveforms coincide:

Table 1: **Uniqueness of the recall cycle.** Phase-aligned comparisons over all 100 stored-pattern initial conditions in the reference network. $d_{\max}$ is the largest distance from the reference waveform; for scale, the waveform itself sweeps overlap values of order one, so the largest residual is below 1% of the signal.

λ	starts	$\bar{T}/t_0$	$d_{\max}$
0.330	100	1869.4	5.3×10^{-3}
0.350	100	1633.5	9.2×10^{-3}

Within this resolution, all 100 memory initial conditions converge to one and the same periodic orbit at each tested λ . The recall cycle is a genuinely global attractor of the memory sequence, though not necessarily the only attractor in the full state space (Sec. 6).

5.3 Exact delay-induced sequential recall

Below λ_* , motion along the forward connection is a transient that ends at the terminal node. At the fold the node and saddle annihilate; above it the same closed route contains no equilibrium and becomes the stable limit cycle of Table 1. Unrolled over pattern index, the orbit is a traveling front whose leading overlap advances from μ to $\mu + 1$ and crosses the cyclic seam from P back to 1; the ordering is verified memory by memory.

The recall it performs is sharper than the geometry of the dying states would suggest. Close

to onset, at $\lambda = 0.330$, the median analog peak over one tour is 0.992 (smallest 0.978), and the binary readout is perfect:

$$b_\mu(t_\mu^{\text{peak}}) = 1.000 \quad \text{for every } \mu = 1, \dots, P, \quad (30)$$

the sign of every neuron reproducing the corresponding stored pattern at that pattern’s own peak. The refined peak-to-peak period at $\lambda = 0.330$ is $1869 t_0$; an independent binary-state classification of the same near-onset cycle finds an exact stored pattern held during 47% of the tour, a fraction that rises to 85% at $\lambda = 0.90$, where the tour shortens to about $1088 t_0$.

Exactness is a puzzle at first sight. The static fixed points from which the orbit inherits its route die as strongly tilted mixtures (successor ratios up to 0.49, Sec. 2.3); why does the moving orbit visit *pure* patterns? The resolution lies in the delay. A fixed point feels its own delayed image ($\mathbf{a}(t - \tau) = \mathbf{a}(t)$), so the directed drive $\lambda S\mathbf{m}$ pushes it permanently toward its successor: the static tilt is obligatory. On the cycle, causality reverses the geometry. When the front peaks at memory μ , the delayed state still carries the previous reign, $\mu - 1$; the delayed drive $\lambda S\mathbf{m}[\mathbf{a}(t - \tau)]$ then points toward μ itself, so instantaneous and delayed fields align and the state purifies toward the exact pattern. Each relay is a consolidation phase (delayed image = $\mu - 1$, drive aligned, duration $\simeq \tau$) followed by a push phase (delayed image = μ , drive tilted, escape in t_{esc}) - which is also how the clock law Eq. (29) arises mechanistically. The tilted mixtures are visited only in motion, except at the critical bond, where the orbit creeps along the fold ghost: at $\lambda = 0.330$ the velocity minimum of the critical relay coincides with the stored fold state to 6×10^{-3} and its dwell time is 2.3 times the median [Fig. 8(e)].

These measurements separate the two roles of the quenched ensemble in the temporal code. The topology of the unique orbit fixes the recall order; the local saddle-node geometry sets the dwell-time distribution, with most of the near-critical period accumulating at the ghost of the single extreme memory. Extreme-value selection thus shapes both the onset of the cycle and its rhythm near onset. We turn next to the regime the cycle leaves behind: the window between the two transitions.

6 Stationary hyperchaos between the two transitions

The interval between a typical threshold and λ_* is a phase with its own dynamics. Most static memories are gone, the survivors retain only small, anisotropic basins, and the terminal barrier still forbids the asymptotic recall cycle. What fills the gap, under the sampled initial-condition ensemble, is a bounded hyperchaotic attractor itinerating among the memory lobes.

6.1 Coexistence with residual retrieval states

At the reference load, 250 initial conditions per λ were drawn from three prescribed families (perturbed single memories, mixtures of consecutive memories, small random states) and classified after $300 t_0$ as stationary retrieval state, ordered moving, irregular, or other. This finite-time classification is deliberately separate from the asymptotic Lyapunov analysis below: below λ_* , a “moving” label can still be a long transit that eventually reaches the terminal node.

At $\lambda = 0.31$ the fractions are 0.052 stationary retrieval states (of which 0.044 on the terminal branch—by then the longest-lived survivor), 0.47 moving and 0.47 irregular; at $\lambda = 0.325$ they are 0.048 (all terminal), 0.55 and 0.40. The residual fixed points capture only a few percent of the sampled ensemble: the window belongs to the irregular flow. Above depinning the hierarchy inverts: at $\lambda = 0.40$ and 0.90 the ordered-cycle fractions are about 80%, with irregular trajectories

Chaotic recall is robust and high-dimensional

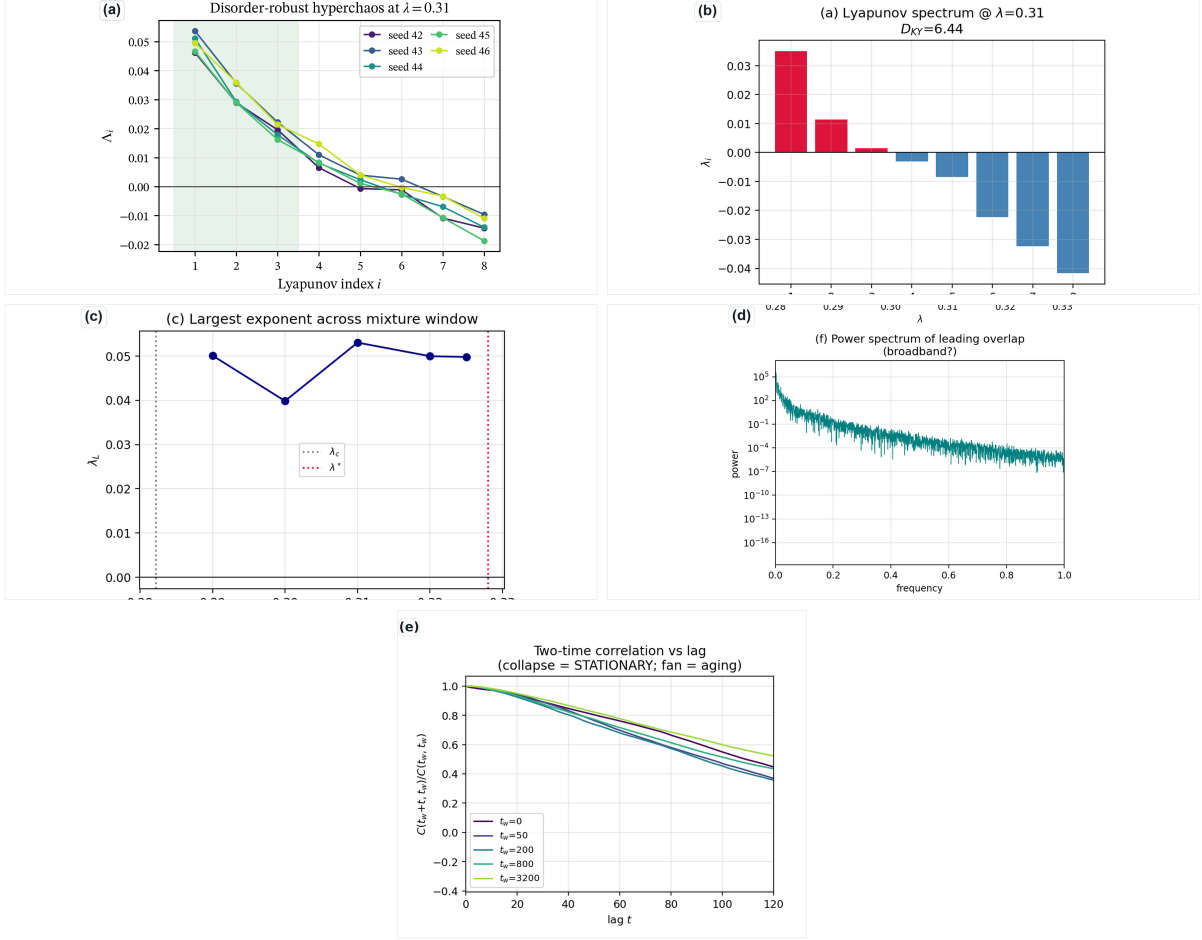


Figure 9: **Hyperchaos in the finite-size bifurcation window.** (a) At $\lambda = 0.31$, the first three Lyapunov exponents are positive in five disorder realizations (refined $\Delta t = 0.005 t_0$ values). (b) The first eight exponents of the reference trajectory give three positive directions and $D_{KY} \simeq 6.4$. (c) The largest exponent remains positive throughout the resolved interval between a typical fold and λ_* . (d) Power spectrum of the leading-overlap time series: broadband, with no discrete periodic comb (see text for the precise definition). (e) Stationarity of the chaos: two-time correlations $C(t_w + t, t_w)/C(t_w, t_w)$ superpose as functions of the lag for waiting times $t_w = 0$ –3200, with no t/t_w aging collapse (Sec. 6.3). $(N, P, \beta, \tau) = (2000, 100, 20, 10)$; panels (b–e) use seed 42.

persisting. The cycle is a robust attractor but does not exhaust the basin structure of this finite network. These fractions are properties of the stated sampling families, not invariant phase-space volumes; their role is to establish coexistence.

6.2 The hyperchaotic strange attractor

The irregular flow is deterministic chaos of moderate dimension, and more than one direction of it. In the reference sample the largest finite-time Lyapounov exponent is positive at all five resolved points of $0.29 \leq \lambda \leq 0.32$, between 0.040 and $0.053 t_0^{-1}$; at $\lambda = 0.31$ the trajectory remains nonstationary over $T = 3000 t_0$ (about 150 inverse-Lyapunov times) with largest exponent

converging to $0.048 t_0^{-1}$. An eight-direction tangent computation gives the representative spectrum

$$(0.035, 0.011, 0.0015, -0.003, -0.008, -0.022, -0.032, -0.042) t_0^{-1}, \quad (31)$$

in which the near-zero third value is the finite-time estimator of the exact neutral flow direction; the Kaplan–Yorke interpolation yields an effective dimension $D_{KY} \simeq 6.4$. This is a finite-window characterization of one attractor—an order of magnitude for the dimension, not a converged extensive quantity, and certainly not extensivity in N .

Panel (d) of Fig. 9 characterizes the temporal signature. The plotted quantity is the power spectrum (the squared modulus of the discrete Fourier transform, after mean subtraction and Hann windowing) of the single overlap time series $a_{\mu_0}(t)$, where μ_0 is the pattern with the largest time-averaged $|a_\mu|$ on the recorded chaotic trajectory at $\lambda = 0.31$. A periodic or quasi-periodic signal would concentrate its power in a discrete comb of lines at the fundamental frequencies and their combinations; the measured spectrum is instead a continuous broadband profile, which excludes a noisy periodic orbit and is the standard temporal fingerprint of deterministic chaos.

The hyperchaos diagnosis replicates across disorder: the attractor at $(\alpha, \lambda, \tau) = (0.05, 0.31, 10)$ is hyperchaotic in all five tested realizations.

6.3 Stationarity: no aging on the resolved windows

The broadband spectrum already excludes a noisy periodic orbit. A subtler alternative—a slow glassy relaxation pretending to be an attractor—is excluded by the two-time correlations of Fig. 9(e), measured at waiting times $t_w \in \{0, 50, 200, 800, 3200\}$. At $\alpha = 0.05$ the correlation curves collapse as functions of the lag: the total spread at lag $20 t_0$ is 0.009 on a normalized correlation of order one (i.e. below 1%), the relative long-wait variation is 0.002, and the decorrelation time is about $2 t_0$; no t/t_w collapse characteristic of aging is observed. The same protocol gives lag-20 spreads of 0.021 at $\alpha = 0.08$ and 0.025 at $\alpha = 0.10$ —still an order of magnitude below the conventional aging threshold of 0.1—with smaller usable ensembles at higher load. The analogy with spin-glass dynamics thus stops at the geometry of the landscape: temporally, the window is a stationary chaotic state sampling a well-defined invariant (SRB-like) measure, not an aging one.

Finally, the window itself is a finite-size object: by Eq. (25), the parameter interval supporting this regime vanishes in the thermodynamic limit. A hyperchaotic phase that exists because the system is finite is a striking expression of the intra-sample mechanism; it also suggests two ways to destroy that mechanism, which the next section carries out.

7 What the mechanism requires: finite delay and discrete barriers

Two logically distinct ingredients underlie the bulk-to-extreme scenario: a propagation time long enough for a localized front to form, and a discrete, heterogeneous set of memory barriers on which that front can pin. Each can be removed independently. Shortening the delay removes the first; correlating successive memories relocates or removes the second. These perturbations are therefore tests of the mechanism itself.

7.1 Short delays delocalize the recall front

At $\lambda = 0.90$ and long delay, the orbit is a succession of narrow, essentially single-memory relays. Reducing τ does not merely accelerate this orbit; it changes its nature. At zero delay the relay

A finite delay synchronizes the propagating recall front

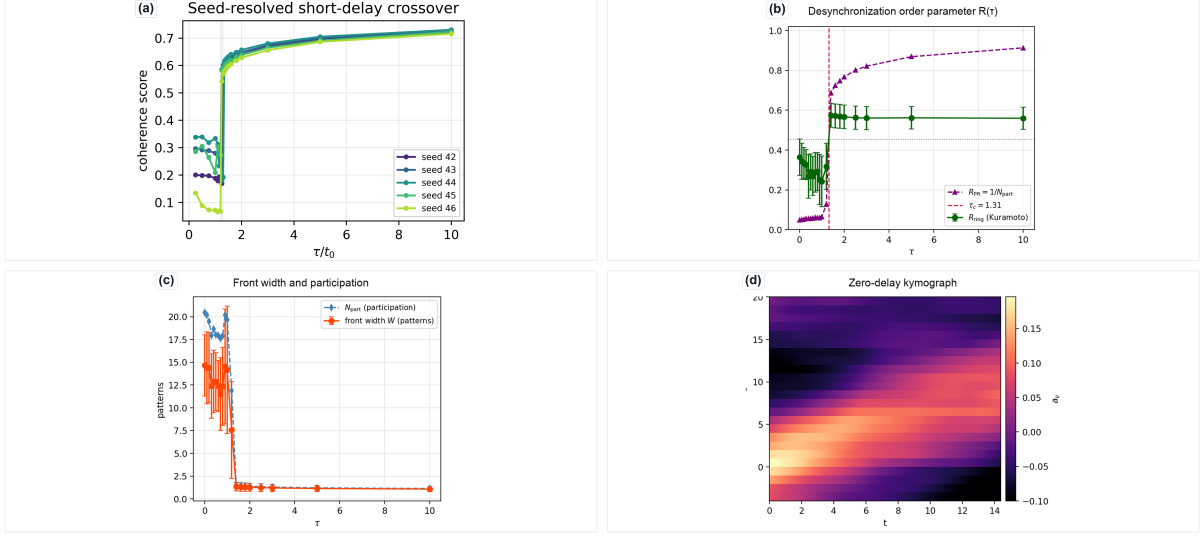


Figure 10: **A finite delay synchronizes the propagating recall front.** (a) A seed-resolved coherence score jumps near $\tau \simeq 1.3$ in all five realizations. (b) Participation-ratio and circular-order diagnostics place the reference crossover at $\tau_c = 1.31$. (c) The number of participating patterns and the front width collapse from a broad packet to an approximately one-pattern front. (d) At zero delay, the overlap kymograph shows the broad desynchronized packet. $(N, P, \alpha, \beta, \lambda) = (2000, 100, 0.05, 20, 0.9)$.

interval is $T_1 = 1.9 \pm 1.2$ (a spread comparable to the mean: there is no clock), the active front spans 15 pattern indices, its largest analog overlap is only 0.16, and its participation number is 21. At $\tau = 1.4$ these numbers become 2.15 ± 0.01 (a 0.5% jitter: a clock), 1.3, 0.67 and 1.5: a broad multi-memory packet has condensed into a reproducible one-memory front (Fig. 10).

The condensation is sharp and disorder robust. The reference crossover sits at $\tau_c \simeq 1.31$, and independent sigmoidal fits of a coherence statistic for seeds 42–46 give midpoints

$$\tau_c = 1.28, 1.24, 1.34, 1.23, 1.23, \quad (32)$$

i.e. a seed-to-seed spread of a few percent around a value of order the neuronal relaxation time $t_0 = 1$. The clock law changes at the same crossover:

$$T_1 \simeq \begin{cases} 0.52 \tau + 1.73, & \tau \lesssim 1, \\ \tau + 0.815, & \tau \gtrsim 2. \end{cases} \quad (33)$$

Below the crossover, relay times are broadly distributed and no single bottleneck dominates; above it, the additive pacemaker clock of Eq. (29) is recovered. Raising λ to 0.99 does not restore the narrow front at short delay: the limiting resource is propagation time, not directed coupling strength.

This clarifies the status of the terminal SNIC at small delay. The static equilibria and their folds do not move (they are τ -independent) but the moving invariant set that must reach them does. Once the front is delocalized, no single bond dominates the arrest: the extreme statistics on which the clean one-bottleneck depinning relies collapse, and the ordered cycle merges into an irregular moving state instead.

7.2 A correlated Markov seam relocates the extreme barrier

To manipulate the barriers themselves, we correlate successive memories through

$$\xi_i^{\mu+1} = \xi_i^\mu s_i^\mu, \quad \Pr(s_i^\mu = -1) = \frac{1-c}{2}, \quad (34)$$

so consecutive patterns overlap by c on average - a “video” of correlated frames. Closing a finite Markov chain into a ring, however, leaves one seam whose overlap is uncorrelated. This topological defect becomes dynamically decisive [Fig. 11(a,b)].

At $N = 10^4$, $P = 500$, $c = 0.2$, the median bulk threshold collapses to 0.062 as similar neighbors pull each memory toward its successor far earlier, while the seam threshold stays at 0.169, by far the largest in the sample. Extreme selection survives, but the selected barrier is no longer a statistical fluctuation: it is the architectural defect, exactly as the bond-disorder predictor of Sec. 2.3 anticipates (aligned neighbors mean low thresholds; the one uncorrelated bond keeps the high one). At stronger correlation the localized states themselves degrade. Correlation has thus not merely shifted the mean threshold; it has relocated the extreme to a geometrically distinguished link, and eventually erased the localized states on which the barrier ensemble was defined.

7.3 Smooth periodic correlations remove pinning altogether

Is the seam essential, or the correlation? To separate them, each neuron’s pattern sequence is generated by thresholding a smooth Gaussian Fourier process on the circle: exactly periodic, hence seam-free, with smoothness set by the number of modes. At nearest-neighbor overlap $q_1 \simeq 0.54$ and $N = 10^4$, only 30 of 500 initialized memory branches retain an identifiable localized fixed point, and at stronger smoothness almost none do. Yet the dynamics is impeccable: forward fraction 1.000 at every tested smoothness, and no pinned state down to $\lambda = 0.20$, far below every threshold of the uncorrelated problem. With the discrete barriers melted into a smooth quasi-continuous ring, there is nothing left to pin on, and the depinning transition itself disappears [Fig. 11(c,d)].

The Markov and smooth constructions thus produce opposite extreme-value fates from the same nominal ingredient, increased local similarity: a seam concentrates pinning into one exceptional link, while homogeneous periodic correlation removes exceptional links altogether. Together with the short-delay experiment, this isolates the minimal requirements of the mechanism: finite propagation time to create a localized traveling front, and quenched heterogeneity to supply discrete, distinguishable barriers with a well-defined extreme. Remove either and the bulk-to-extreme scenario no longer organizes the dynamics. The explicit point delay, however, is only one way to provide the propagation time; we close by showing that a perhaps more biologically realistic hidden architectural depth serves equally well.

8 Implicit delay from Markovian depth, and general implications

The discrete delay of Eq. (2) makes the relay clock transparent, but no biological architecture implements a literal delay operator. A larger Markovian system, even if more complex than simple state-storing, can hold the required history in hidden dynamical variables instead. We verify this with a layered recurrent network, then separate what is model-specific in our results from the general finite-size mechanism they instantiate.

Correlated memories: one construction concentrates pinning, the other removes it

(a,b) Markov flip chain closed into a ring: one uncorrelated seam. (c,d) exactly periodic circular Gaussian process: no seam. $N = 10^4$, $P = 500$, $\beta = 20$, seed 42.

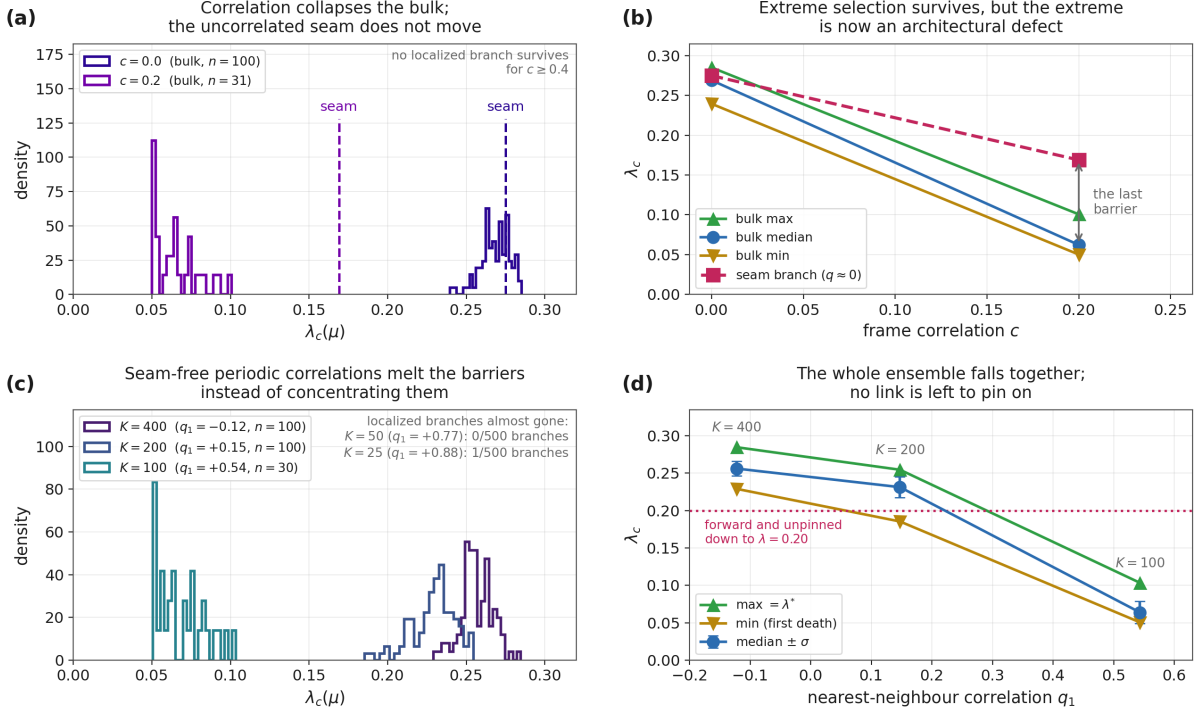


Figure 11: **Correlated memories: one construction concentrates pinning, the other removes it.** (a,b) Markov flip chain closed into a ring, Eq. (34). (a) The bulk threshold distribution moves downward as the frame correlation c increases, while the dashed line marks the uncorrelated closing seam, which does not move. (b) Bulk minimum, median and maximum versus c , against the seam branch: at $c = 0.2$ the seam stays at 0.169 while the bulk median falls to 0.062, so the surviving extreme is no longer a statistical fluctuation but an architectural defect. (c,d) Seam-free circular Gaussian-process patterns with K latent Fourier modes. (c) Threshold distributions versus smoothness: as the ring becomes smooth the localized primary branches disappear, 30 of 500 surviving at $q_1 \simeq 0.54$ and essentially none beyond. (d) Minimum, median and maximum versus the nearest-neighbour correlation q_1 ; the whole ensemble falls together, leaving no exceptional link, and the dynamics remains forward and unpinned down to $\lambda = 0.20$, the lowest value tested. $(N, P, \beta) = (10^4, 500, 20)$, seed 42.

8.1 A layered architecture with implicit memory

Let $U_k \in \mathbb{R}^N$ be the activity of layer $k = 0, \dots, K_{\text{lay}} - 1$ and $g(U) = \tanh(\beta U)$. Intermediate layers carry local autoassociative recurrence and receive the preceding layer,

$$t_0 \dot{U}_k = -U_k + Jg(U_k) + \lambda g(U_{k-1}), \quad 1 \leq k < K_{\text{lay}}, \quad (35)$$

and the ring closes through the cyclic successor map,

$$t_0 \dot{U}_0 = -U_0 + K_{\text{seq}} g(U_{K_{\text{lay}}-1}). \quad (36)$$

This system is an ordinary differential equation, Markovian in the enlarged layer state (Fig. 12). Eliminating the hidden layers, however, leaves the observed layer with a distributed, state-dependent memory kernel - not the point delay of Eq. (2), but a candidate for the same clocking role.

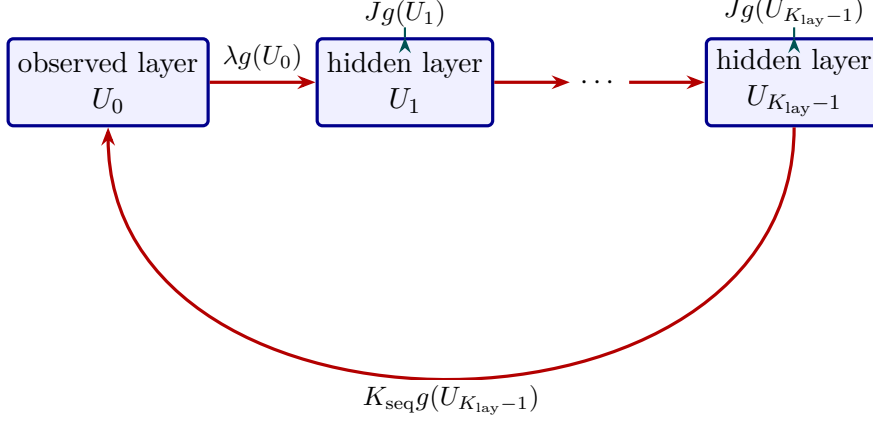


Figure 12: **Markovian architecture generating an implicit delay.** Activity propagates through recurrent hidden layers before the last layer drives the cyclic successor of the observed layer. The full state is Markovian; the observed layer alone inherits a distributed history dependence from depth.

8.2 Depth sets the relay clock

Simulations at $N = 500$, $P = 25$, $\beta = 20$, for $K_{\text{lay}} = 4, 6, 8, 11$, couplings $\lambda = 1, 2, 4$ and five disorder seeds give a recall period linear in the number of intermediate propagation steps,

$$T_1(K_{\text{lay}}, \lambda) = a(\lambda) + b(\lambda) (K_{\text{lay}} - 1), \quad (37)$$

with $R^2 \geq 0.999$ in every fit. The slopes (mean $\pm$ SD across seeds) are

$$\begin{aligned} b(1) &= 1.67 \pm 0.08, \\ b(2) &= 1.17 \pm 0.02, \\ b(4) &= 0.94 \pm 0.01, \end{aligned} \quad (38)$$

so the seed-to-seed spread is at most 5% of the slope itself. Two independent estimators of the hidden propagation time - the cross-correlation shift between layers and the delay between matched relay events - reproduce the slopes to within 3.4% in the worst case. Depth therefore supplies a reproducible effective delay quantum per layer.

What is the value of that delay quantum, and in particular its strong-coupling limit? Intuition suggests that at very large λ the Hebbian term becomes irrelevant and the chain reduces to $K_{\text{lay}} - 1$ layers that simply store and pass the state, i.e. an effective delay exactly proportional to depth, which is indeed what Eq. (37) expresses. The naive version of this intuition, however, would set the per-layer quantum equal to the relaxation time t_0 ($\tau_{\text{eff}} = K_{\text{lay}} - 1$ in our units). The data refute this quantitatively. On a dense coupling grid $\lambda \in [1, 8]$ (12 values $\times$ 4 depths $\times$ 5 seeds, all with $R^2 \geq 0.999$), the quantum decreases strictly with λ and is best described by

$$b(\lambda) \simeq 0.75 + 0.91 \lambda^{-1.1}, \quad (39)$$

with per-seed plateau values $b_\infty \in [0.73, 0.79]$ and exponents $p \in [1.0, 1.3]$. The per-layer delay therefore saturates near $0.75 t_0$ at strong coupling rather than approaching t_0 .

A minimal relay argument shows why a sub-unit plateau is expected. At large λ each hidden layer obeys $t_0 \dot{u} \simeq -u + \lambda \tanh(\beta u_{\text{prev}})$, and both its steady state and its drive scale with λ . When

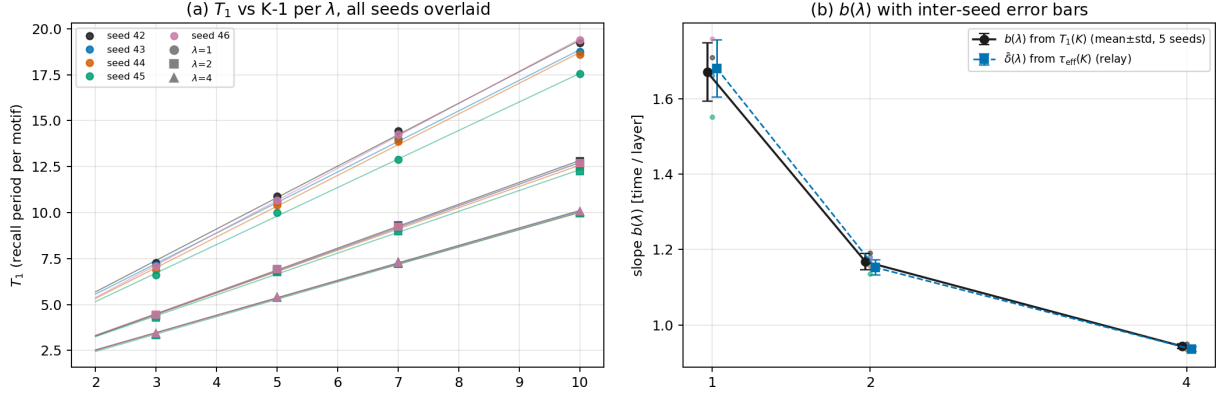


Figure 13: **Depth generates an effective neural delay.** Across five disorder realizations the recall period is linear in $K_{\text{lay}} - 1$; slopes agree with independently measured inter-layer relay and cross-correlation delays. The per-layer quantum decreases with the propagation coupling λ and saturates near $0.75 t_0$, as expected for a dynamical switching time rather than a fixed discrete delay.

the input flips sign, the layer relaxes from $\simeq -\lambda$ toward $+\lambda$, so it crosses zero (and its saturated output flips) after $t_0 \ln 2 \simeq 0.69 t_0$ independently of λ , because the starting point and the target grow together. The measured plateau $b_\infty \approx 0.75$ sits close to, and slightly above, this switching bound, the residual excess being consistent with the finite flip time of the upstream layer and the neglected Hebbian pull. We flag this argument as an interpretation: what is established numerically is the linear depth law and the saturating quantum of Eq. (39), at the stated sizes ($N = 500$ – 2000 with $P = \alpha N$ covarying, couplings $\lambda \leq 8$); no formal $\lambda \rightarrow \infty$ or $N \rightarrow \infty$ statement is demonstrated.

On the coarse regime grid, all five seeds are stationary for $\lambda \leq 0.5$ and support an ordered wave for $\lambda \geq 1$, with the onset refined to the bracket $(0.64, 0.70]$ at $K_{\text{lay}} = 6$ and a narrow irregular-switching band just below it. The essential point is Eq. (37): an explicit point delay is unnecessary; hidden dynamical depth provides the propagation memory that the mechanism requires.

9 Conclusion: from one neural network to a general finite-size mechanism

The microscopic conclusions of this article are specific: in the delayed non-reciprocal Hopfield network, every memory-connected equilibrium carries a quenched fold threshold. Within one sample these thresholds form a quasi-i.i.d., near-Gaussian, position-scale family whose standardized shape is universal across loads. The bulk of the law ends extensive static recall, while its upper extreme selects the bottleneck memory in all tested disorder realizations and predicts the operational depinning boundary (at 10^{-3} resolution), in quantitative agreement with Gaussian extreme-value theory. The terminal event is a SNIC on a directly closed invariant recall circle reached from every stored pattern, which performs exact binary recall at every memory peak and ticks with the additive clock $T_1 = \tau + t_{\text{esc}}$. Between the two boundaries lives a stationary hyperchaotic attractor. Ablation experiments identify two necessary ingredients to generalize the phenomena: a delay-localized traveling front and discrete heterogeneous barriers, while a more biologically feasible layered Markovian architecture shows that the delay itself may be implicit.

We argue this work may teach us something more general about finite disordered systems.

When one realization contains many local states that a clean or averaged description treats as equivalent, quenched disorder can split their bifurcations into an internal ensemble. A collective observable that samples a typical local state and one that requires completion through *all* local states will then select different order statistics of the same law, and the resulting finite-size phase boundaries are distinct physical events, possibly with an entire dynamical phase between them, rather than competing estimates of one transition. If the internal law narrows as size grows, as measured here, the boundaries merge and the thermodynamic description retains a single critical point that hides the selection mechanism. Finite size is then not a defect of the description but the regime in which the mechanism is observable at all.

The ingredients of this construction are not specific to neural networks: they are a frustrated disordered landscape, a non-equilibrium drive, and quenched heterogeneity of local thresholds. We therefore propose it as an operational diagnostic: resolve the local bifurcations within one sample, measure their one-point law, correlations and finite-size narrowing, and ask which order statistic each collective observable follows. In the present system, that program explains the birth of sequential memory recall as resulting from the statistical mechanics of quenched bifurcations.